A ABAQUS explicit phase field frictional heating implementation using a VUMAT-VUMATHT subroutine

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Abstract

Documentation that accompanies the VUMAT–VUMATHT subroutine provided to implement the phase field frictional heating method in ABAQUS/Explicit, extendable for adding new fracture driving forces, without the need for user elements, and easily extensible to any general thermo-mechanical simulation — not limited to frictional thermo-mechanical coupling. The code can be downloaded from https://github.com/lalit550/Phase-field-frictional-heating-vumat-vumatht. If using these codes for research or industrial purposes, please cite the attached articles.

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1 Introduction

Phase field fracture has become a widely adopted approach for modelling crack nucleation and propagation due to its non-local, regularised representation of damage, which removes the need to explicitly track discrete crack surfaces. The method recasts a discontinuous crack topology as a smooth scalar field $\phi \in [0,1]$ ($\phi = 0$: intact, $\phi = 1$: fully broken), governed by a diffusion-type evolution law coupled to the mechanical balance equations (1). For a solid with toughness G_c and length scale ℓ, the phase field evolves as

$$\frac{\phi}{\ell^2} - \nabla^2\phi \;=\; -\,\frac{2(1-\phi)}{G_c\,\ell}\,\mathcal{H}, \tag{1}$$

where $\mathcal{H}$ is the crack driving (history) force, enforced to be monotonically non-decreasing to satisfy irreversibility, $\mathcal{H} = \max_{\tau \le t} \psi^+(\tau)$, with ψ^+ the tensile part of the strain energy density obtained from a spectral or volumetric-deviatoric decomposition of the strain tensor. The present implementation retains the Miehe-type spectral split (2) as the reference/comparison model, while allowing the degradation function $g(\phi) = (1-\phi)^2 + k$ (with $k \ll 1$ a residual stiffness parameter) to act on ψ^+ only, preserving the compressive response of the material.

In addition to crack-tip damage evolution, frictional heating along sliding crack faces is an important secondary dissipation mechanism, particularly relevant to impact and shock-loading problems where post-fracture surfaces continue to slide and generate heat. In discrete crack methods (cohesive elements, XFEM, or explicit crack insertion), computing this contribution requires an explicit crack geometry to be inserted *a priori*, tracking of contact pairs along that geometry, and an explicit definition of the sliding surface and its normal/tangential directions. This is both computationally expensive and, because it depends on how the crack is discretised and constrained, frequently under-predicts the true frictional dissipation.

Phase field fracture offers a natural alternative: the diffuse crack representation already carries the local crack orientation (through $\nabla\phi$) and an implicit measure of crack surface area (through the crack surface density functional γ), so no additional crack geometry or contact pair needs to be defined. A classical Coulomb friction law can then be recovered by weighting the point-wise frictional power by the local crack surface density, converting a surface (per-unit-area) heat generation rate into a volumetric one:

$$\dot{q}_{\text{fric}} \;=\; \mu\,\langle -t_n \rangle\, v_s\, \gamma(\phi, \nabla\phi), \qquad \gamma(\phi,\nabla\phi) = \frac{\phi^2}{2\ell} + \frac{\ell}{2}\,|\nabla\phi|^2, \tag{2}$$

where μ is the friction coefficient, $\langle -t_n \rangle$ is the compressive part of the traction resolved onto the local crack normal $\mathbf{n} = -\nabla\phi/|\nabla\phi|$, and v_s is the local sliding (slip) velocity. This formulation follows the frictional heat generation concept of grilli et al. (3), here re-derived for an explicit-dynamics, VUMAT/VUMATHT setting.

A key practical difficulty in discrete methods is the calculation of a physically admissible slip velocity, which typically requires constraining or capping the result to avoid spurious values at contact initiation or under large mesh distortion. In the present implementation, the slip velocity is instead obtained directly from the local velocity gradient $\mathbf{L}$, computed from the deformation gradient rate,

$$\mathbf{L} = \dot{\mathbf{F}}\,\mathbf{F}^{-1}, \qquad v_s = \big|\,\mathbf{t}\cdot(\mathbf{L}\,\mathbf{n})\,\big|\;\ell, \tag{3}$$

with $\mathbf{t}$ the in-plane tangent to the crack ($\mathbf{t} \perp \mathbf{n}$). To avoid the ill-conditioning of Eq. (3) under severe element distortion (where $\det \mathbf{F}$ departs strongly from unity), a bounded fallback based directly on the incremental strain and spin ($\Delta\varepsilon/\Delta t$, $\Delta\omega/\Delta t$) is used, which avoids the need for an externally imposed velocity cap while remaining consistent with the incremental kinematics already available inside the VUMAT.

The remainder of this documentation describes the VUMAT/VUMATHT architecture (Section 2), the SDV communication scheme between the mechanical and thermal-analogy subroutines (Section 2), usage instructions (Section 3), and a representative example (Section 4). Two VUMAT–VUMATHT code are provided alongside this documentation, both following the architecture described above but differing in the strain-energy split used to drive $\mathcal{H}$: one implements the Miehe-type spectral split, used here for comparison purposes, and the other implements the deviatoric tensile/compressive split of Eq. (4). Both variants include the frictional heating extension and share the same `pf_shared` communication scheme.

2 ABAQUS implementation

The thermal analogy is exploited so that the phase field ϕ is represented by the temperature degree of freedom T in a *Coupled temperature-displacement* analysis, with unit "conductivity" and a user-defined internal heat generation term. Since ABAQUS/Explicit does not support a coupled UMAT+HETVAL pair, the framework instead uses two independent user subroutines that communicate through shared, element-indexed Fortran module arrays:

- **VUMAT** — solves the mechanical (displacement) problem: computes strain, performs the strain-energy decomposition, updates the degraded Cauchy stress, evaluates the crack driving force $\mathcal{H}$, and computes the frictional heat generation rate $\dot{q}_{\text{fric}}$ then solve for real or hotspot temperature.
- **VUMATHT** — solves the phase field (temperature-analogy) problem: reads $\mathcal{H}$ computed by the VUMAT, evaluates the volumetric heat generation term R_{src} and its capacity term, and writes the updated phase field ϕ and its gradient $\nabla\phi$ back for the VUMAT to consume in the next increment for calculations for frictional heat source so the real temperature.

2.1 Inter-subroutine communication

Since VUMAT and VUMATHT are called independently by ABAQUS/Explicit for a given element (and not necessarily in a fixed order relative to each other within an increment), a shared Fortran module (`pf_shared`) is used, exposing package-level arrays indexed by a local element number `locnum`, obtained via `VGETPARTINFO` in both subroutine:

```
module pf_shared
  real*8 H_store(MAX_ELEM)     ! crack driving force H, written by VUMAT
  real*8 phi_store(MAX_ELEM)   ! phase field phi, written by VUMATHT
  real*8 Tgrad(MAX_ELEM,3)     ! grad(phi), written by VUMATHT
end module
```

This lagged (staggered-by-one-increment) exchange is consistent with the explicit time integration scheme itself, in which all field quantities are already updated explicitly from values known at the start of the increment; the module arrays therefore introduce no additional approximation beyond that already inherent to the explicit scheme.

2.2 Strain energy decomposition and stress degradation (VUMAT)

At each material point, the total strain is accumulated from SDVs and used to construct the undamaged principal stresses. Two split strategies are supported: (i) a Miehe-type spectral split of the full strain tensor, used here as the reference/comparison model, and (ii) a volumetric tensile/compressive split combined with an eigen-decomposition of the *deviatoric* strain, so that only the tensile volumetric and tensile deviatoric parts contribute to the crack driving energy,

$$\psi^+ = \tfrac{1}{2}K\langle \operatorname{tr}\varepsilon\rangle_+^2 + \mu\,\mathbf{e}^+ : \mathbf{e}^+, \qquad \psi^- = \tfrac{1}{2}K\langle \operatorname{tr}\varepsilon\rangle_-^2 + \mu\,\mathbf{e}^- : \mathbf{e}^-, \tag{4}$$

where $\mathbf{e}^{\pm}$ are the positive/negative projections of the deviatoric strain eigenvalues, rotated back to the global frame. The degraded stress is then assembled as $\boldsymbol{\sigma} = \boldsymbol{\sigma}^- + g(\phi)\,\boldsymbol{\sigma}^+$, with $g(\phi) = (1-\phi)^2 + k$, so that compressive/undamaged contributions bypass degradation entirely because we have to calculate frictional heat by this compression. The crack driving force is updated under the irreversibility constraint $\mathcal{H} = \max(\mathcal{H}_{\text{old}}, \psi^+)$ and written to `H_store`.

2.3 Phase field evolution (VUMATHT)

The temperature-analogy source term is defined directly from the heat-conduction/phase-field correspondence, with unit conductivity ($k = 1$) and the volumetric source

$$R_{\text{src}} = \frac{\phi}{\ell^2} + \frac{g'(\phi)}{\ell\, G_c}\,\mathcal{H}, \qquad g'(\phi) = -2(1-\phi), \tag{5}$$

computed using the value of $\mathcal{H}$ retrieved from `H_store` at the corresponding `locnum`. The flux is set as $\mathbf{f} = -\nabla\phi$ (mobility = 1), and the thermal capacity term $\partial(\Delta U)/\partial T = C_p$ is returned as the (constant) specific heat analogue. After each increment, VUMATHT writes the updated ϕ and $\nabla\phi$ back into `phi_store` and `Tgrad`, making them available to the VUMAT call of the following increment so that VUMAT can calculate frictional heat source.

2.4 Frictional heating

The frictional heat generation rate is evaluated inside the VUMAT, using the crack normal $\mathbf{n} = -\nabla\phi/|\nabla\phi|$ (from `Tgrad`), the local velocity gradient $\mathbf{L}$ (with a bounded incremental-strain fallback when $\det\mathbf{F}$ departs from $[0.80, 1.25]$), and the compressive normal traction $\langle -t_n\rangle$ computed from the current degraded stress. The volumetric frictional power is

$$\dot{q}_{\text{fric}} = \mu\,\langle -t_n\rangle\,\gamma(\phi, \nabla\phi)\,v_s, \tag{6}$$

reported through `enerInelasNew` for global energy balance checks and for calculation for real temprature rise. This term is independent of, and additive to, the crack-propagation heat source obtained from Eq. (5), allowing the two dissipation mechanisms to be enabled, weighted, or studied separately ,Eq. (5) is thermal balance for phase field fracture calculation while Eq. (6) is thermal balance for real hotspot temperature calculation .

2.5 State variable (SDV) layout

SDV	Contents
1–4	Accumulated strain components (or stress components, split-dependent)
7	Crack driving force $\mathcal{H}$ (mirror of `H_store`)
30–32	Slip velocity v_s, compressive traction, crack density weight γ

3 Usage instructions

The same overall process as a standard Abaqus/Explicit model is followed, with the additions below. Two overlapping meshes coexist of the same geometry in a assembly: a *mechanical* part carrying the real material properties (density, conductivity, specific heat — used by `VUMAT` and the genuine thermodynamic heat balance) and a *phase-field clone* part carrying dummy thermal properties that make the temperature DOF solve the phase-field analogy equation (used by `VUMATHT`).

1. **Mechanical material (VUMAT).** Defined as a *User Material*(user material type = Mechanical) with 4 constants:

Parameter	`PROPS`
Young's modulus E	`PROPS(1)`
Poisson's ratio ν	`PROPS(2)`
Phase field length scale ℓ	`PROPS(3)`
Toughness G_c	`PROPS(4)`

   This block also carries the real `*Conductivity`, `*Density`, `*Specific Heat` and `*Inelastic Heat Fraction` cards, since it participates in the genuine thermal solution (frictional/inelastic heat generation, real temperature rise). 32 `Depvar` are required (SDV layout as listed in Section 2 but many are unused from sdv13 to 29 which can be either removed or make use for a better visulization of field vartibales).

2. **Phase-field clone material (VUMATHT).** Defined as a *User Material, type=THERMAL* with 2 constants:

Parameter	`PROPS`
Phase-field heat capacity analogue C_p	`PROPS(1)`
Fracture toughness G_c	`PROPS(2)`

   No `*Conductivity` or `*Specific Heat` cards are needed on this material: C_p and G_c are supplied entirely through `PROPS` and consumed directly inside `VUMATHT` via `condEff(km) = mobility` and `dEnerThDTemp(km,1) = Cp`. A dummy `*Elastic` property (modulus E and Poisson's ratio ν) must still be defined on this material, since Abaqus requires an elastic definition to compute the initial stable time increment; these values play no role in the phase-field solution itself.

   The mobility (= 1) and the length scale ℓ are currently hard-coded in the subroutine rather than read from `PROPS`.

3. **Analysis step.** The step is of type *Dynamic, Temperature-displacement, Explicit*, e.g. `*Dynamic Temperature-displacement, Explicit`, with an optional fixed time increment and standard bulk viscosity coefficients (`*Bulk Viscosity`).

4. **Initial conditions.** Two separate `*Initial Conditions, type=TEMPERATURE` cards are required, one per overlapping part. On the *mechanical* part, the real initial temperature of the material is prescribed (e.g. $T_0 = 298$ K), since this part participates in the genuine thermal/energy balance. On the *phase-field clone* part, the temperature DOF represents ϕ, so it must be initialised to $\phi(t = 0) = 0$ everywhere, unless a pre-existing crack is seeded as an elevated initial ϕ in that region.

5. **Boundary/loading conditions.** Applied on the mechanical part. Periodic boundary conditions are imposed on the top and bottom edges. On the left edge, the horizontal displacement is fully constrained (`*Boundary`, `1,1`), while on the right edge a velocity boundary condition is applied in the horizontal direction, driven by an `*Amplitude` definition (`*Boundary, amplitude=AMP-1, type=VELOCITY`, `1,1,` $-v$), prescribing the impact velocity v.

6. **Output requests.** Field output should include `S`, `LE`, `SDV`, `TEMP`, `HFL` and `ER`; nodal output should include `NT` (phase field visualisation), `U`, `V`. `*Restart, write` is recommended given the typically long run times of explicit phase-field simulations.

4 Representative example

As a representative example, the coupled VUMAT–VUMATHT framework is applied to a PBX-9501 microstructure impacted at the mesoscale, used throughout this documentation to illustrate the combined phase-field fracture and frictional-heating response.

4.1 Problem description

The simulated domain is a two-dimensional PBX-9501 microstructure containing HMX crystals embedded in an Estane binder matrix, discretised with the mechanical part (`PART-1`) and its phase-field clone part (`PART-2`) sharing identical nodal coordinates, as described in Section 3. Periodic boundary conditions are applied on the top and bottom edges; the left edge is fully constrained in the horizontal direction, and a velocity boundary condition, driven by a smooth-step amplitude, is applied on the right edge to represent impact loading at $v = 160$ m/s. The mechanical material (`MAT1`/`MAT2`) uses the properties listed in Table 1, with a phase-field length scale $\ell = 0.4$–4 μm matched to the mesh resolution and a friction coefficient $\mu = 0.5$ on crack surfaces.

Table 1: Representative mechanical and phase-field material properties used in the PBX-9501 example.

Property	HMX	Estane
Young's modulus E	2.12×10^{10} Pa	2.98×10^{8} Pa
Poisson's ratio ν	0.277	0.48
Density ρ	1903 kg/m^3	1186 kg/m^3
Fracture toughness G_c	252.5 J/m^2	80.85–192 J/m^2
Length scale ℓ	4 μm	

An initially undamaged phase field ($\phi(t = 0) = 0$) is prescribed on the clone part, and a uniform initial temperature $T_0 = 298$ K is prescribed on the mechanical part. The analysis step is of type *Dynamic, Temperature-displacement, Explicit*, with standard bulk viscosity coefficients, and no mass scaling applied.

4.2 Time integration and temporal resolution

The coupled problem is integrated in time with the explicit scheme of Abaqus/Explicit. The time increment is not prescribed directly by the user; it is the automatic stable increment, evaluated from the element size, density and stiffness on the mechanical mesh and from the effective heat capacity and conductivity on the phase-field (thermal) mesh. No mass scaling was used. For the PBX-9501 impact simulation of this section, the initial increment is $\Delta t \approx 62$ ps, decreasing to roughly 10–30 ps as the mesh deforms near the crack path, so that approximately 6.4×10^3 increments cover the 180 ns simulated at the default time scaling. The frictional heat source and the resulting temperature rise therefore evolve over several thousand increments and are well resolved by the time integration. The data exchange between VUMAT and VUMATHT is staggered by one increment, i.e. of the order of tens of picoseconds.

To verify that the results are insensitive to the time increment, the same simulation was repeated with a time-scaling factor of 0.5 applied to the automatic stable increment, roughly doubling the number of increments (from $\approx 6.4 \times 10^3$ to $\approx 1.1 \times 10^4$) over the same 180 ns window. Figure 1 compares the temperature history of the hottest element predicted with and without this additional time-scaling. The two curves are visually indistinguishable up to ∼130 ns and remain within 3% of each other over the full simulated window, including through the steep temperature rise beyond 140 ns. This confirms that the default, automatically-computed stable time increment already resolves the frictional and crack-propagation heat sources without.

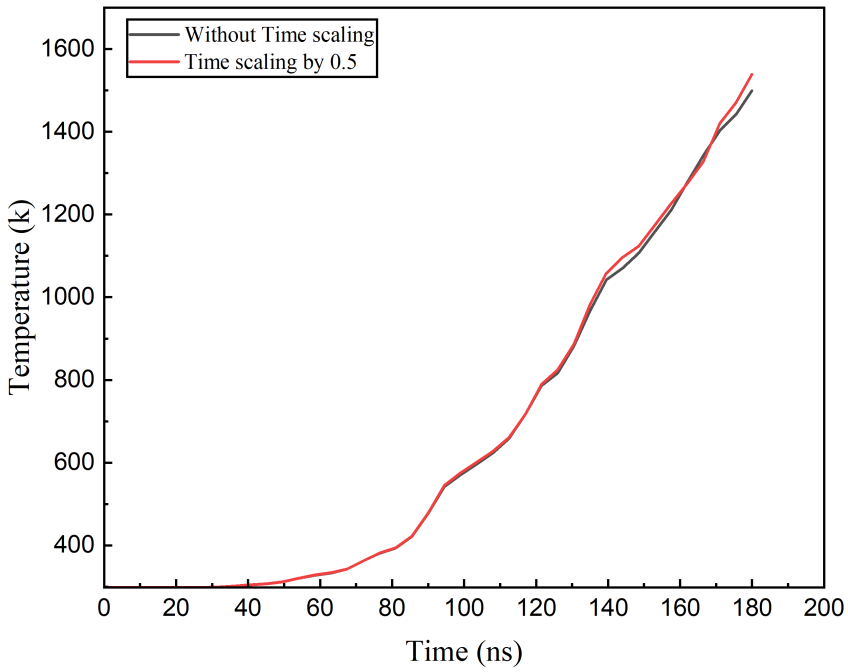


Figure 1: Temperature history of the hottest element in the PBX-9501 representative example, comparing the default automatic stable time increment (black) against a run with the time-scaling factor reduced to 0.5 (red). The two predictions agree to within 3% over the full 180 ns simulated.

A Variational Derivation of the Governing Equations

This appendix summarizes the variational derivation of the coupled momentum balance and phase-field evolution equations used in the present work. The derivation is included for completeness, while the main text focuses on the resulting governing equations.

Variation with respect to the displacement field $\boldsymbol{u}$

The kinetic contribution to the action functional is given by

$$W_k = \frac{1}{2}\int_\Omega \rho\, \dot{\boldsymbol{u}} \cdot \dot{\boldsymbol{u}}\, \mathrm{d}V. \tag{A.1}$$

Its variation with respect to the displacement field yields

$$\delta W_k = \int_\Omega \rho\, \ddot{\boldsymbol{u}} \cdot \delta\boldsymbol{u}\, \mathrm{d}V = -\int_\Omega \rho\, \ddot{\boldsymbol{u}} \cdot \delta\boldsymbol{u}\, \mathrm{d}V, \qquad \delta\boldsymbol{u}\big|_0 = \delta\boldsymbol{u}\big|_T = \boldsymbol{0}. \tag{A.2}$$

The variation of the elastic energy contribution is

$$\delta \int_\Omega \psi_1(\boldsymbol{\varepsilon}, \phi)\, \mathrm{d}V = \int_\Omega \frac{\partial \psi_1}{\partial \boldsymbol{\varepsilon}} : \delta\boldsymbol{\varepsilon}\, \mathrm{d}V + \int_\Omega \frac{\partial \psi_1}{\partial \phi}\, \delta\phi\, \mathrm{d}V. \tag{A.3}$$

Define the degraded Cauchy stress

$$\boldsymbol{\sigma}(\boldsymbol{\varepsilon}, \phi) = \frac{\partial \psi_1}{\partial \boldsymbol{\varepsilon}}. \tag{A.4}$$

Using

$$\int_\Omega \boldsymbol{\sigma} : \delta\boldsymbol{\varepsilon}\, \mathrm{d}V = -\int_\Omega (\boldsymbol{\nabla} \cdot \boldsymbol{\sigma}) \cdot \delta\boldsymbol{u}\, \mathrm{d}V + \int_{\partial\Omega} (\boldsymbol{\sigma} \cdot \boldsymbol{n}) \cdot \delta\boldsymbol{u}\, \mathrm{d}S. \tag{A.5}$$

External variation:

$$\delta\left[\int_\Omega \boldsymbol{b}\cdot\boldsymbol{u}\,\mathrm{d}V+\int_{\partial\Omega_t}\boldsymbol{t}\cdot\boldsymbol{u}\,\mathrm{d}S\right]=\int_\Omega \boldsymbol{b}\cdot\delta\boldsymbol{u}\,\mathrm{d}V+\int_{\partial\Omega_t}\boldsymbol{t}\cdot\delta\boldsymbol{u}\,\mathrm{d}S. \tag{A.6}$$

Collecting terms:

$$\int_\Omega\left(-\rho\,\ddot{\boldsymbol{u}}-\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}-\boldsymbol{b}\right)\cdot\delta\boldsymbol{u}\,\mathrm{d}V+\int_{\partial\Omega}(\boldsymbol{\sigma}\cdot\boldsymbol{n}-\boldsymbol{t})\cdot\delta\boldsymbol{u}\,\mathrm{d}S=0. \tag{A.7}$$

Since $\delta\boldsymbol{u}$ is arbitrary, obtain the momentum balance (strong form)

$$\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}+\boldsymbol{b}=\rho\,\ddot{\boldsymbol{u}},\quad\text{in }\Omega_u, \tag{A.8}$$

with traction BC

$$\boldsymbol{\sigma}\cdot\boldsymbol{n}=\boldsymbol{t}\quad\text{on }\partial\Omega_t,\qquad\boldsymbol{u}=\bar{\boldsymbol{u}}\quad\text{on }\partial\Omega_u. \tag{A.9}$$

Variation with respect to ϕ (phase field)

The viscous regularisation contribution associated with the phase-field evolution is

$$W_v=\int_\Omega\frac{1}{2}\eta\,\dot{\phi}^2\,\mathrm{d}V, \tag{A.10}$$

whose variation yields

$$\delta W_v=\int_\Omega\eta\,\dot{\phi}\,\delta\phi\,\mathrm{d}V. \tag{A.11}$$

The variation of the elastic energy with respect to ϕ is

$$\delta\int_\Omega\psi_1(\boldsymbol{\varepsilon},\phi)\,\mathrm{d}V=\int_\Omega\frac{\partial\psi_1}{\partial\phi}\,\delta\phi\,\mathrm{d}V+(\text{terms in }\delta\boldsymbol{u}\text{ already accounted for}). \tag{A.12}$$

The AT2 fracture functional (gradient + local) part is $\dfrac{G_c}{2}\left(\dfrac{\phi^2}{\ell}+\ell\,|\boldsymbol{\nabla}\phi|^2\right)$; its variation is

$$\delta\int_\Omega\frac{G_c}{2}\left[\frac{\phi^2}{\ell}+\ell\,|\boldsymbol{\nabla}\phi|^2\right]\mathrm{d}V=\int_\Omega\frac{G_c}{\ell}\,\phi\,\delta\phi\,\mathrm{d}V+\int_\Omega G_c\,\ell\,\boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\delta\phi\,\mathrm{d}V. \tag{A.13}$$

Integration by parts of the gradient term gives

$$\int_\Omega G_c\,\ell\,\boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\delta\phi\,\mathrm{d}V=-\int_\Omega G_c\,\ell\,\Delta\phi\,\delta\phi\,\mathrm{d}V+\int_{\partial\Omega}G_c\,\ell\,(\boldsymbol{\nabla}\phi\cdot\boldsymbol{n})\,\delta\phi\,\mathrm{d}S. \tag{A.14}$$

Stationarity of the Lagrangian ($\delta\mathcal{L}=0$) requires the viscous dissipation to balance the variations of the stored and fracture energies. Collecting all $\delta\phi$ contributions and requiring the result to vanish for arbitrary $\delta\phi$ in Ω:

$$\int_\Omega\left[\eta\,\dot{\phi}-\frac{\partial\psi_1}{\partial\phi}-\frac{G_c}{\ell}\,\phi+G_c\,\ell\,\Delta\phi\right]\delta\phi\,\mathrm{d}V-\int_{\partial\Omega}G_c\,\ell\,(\boldsymbol{\nabla}\phi\cdot\boldsymbol{n})\,\delta\phi\,\mathrm{d}S=0. \tag{A.15}$$

Because $\delta\phi$ is arbitrary in Ω and on $\partial\Omega$ independently, the phase-field strong-form equation and natural boundary condition are

$$\eta\,\dot{\phi}=\frac{\partial\psi_1}{\partial\phi}+\frac{G_c}{\ell}\,\phi-G_c\,\ell\,\Delta\phi,\qquad\text{in }\Omega, \tag{A.16}$$

$$\boldsymbol{\nabla}\phi\cdot\boldsymbol{n}=0,\qquad\text{on }\partial\Omega. \tag{A.17}$$

To enforce irreversibility of fracture, the elastic driving term $\partial\psi_1/\partial\phi = g'(\phi)\,\psi^+$ is replaced by the history field

$$\mathcal{H}(\boldsymbol{x},t) = \max\{\mathcal{H}(\boldsymbol{x},t^-),\ \psi^+(\boldsymbol{\varepsilon}(\boldsymbol{x},t))\}, \tag{A.18}$$

leading to the final evolution equation used in this work:

$$\eta\,\dot{\phi} = g'(\phi)\,\mathcal{H} + \frac{G_c}{\ell}\,\phi - G_c\,\ell\,\Delta\phi. \tag{A.19}$$

B Tangential Slip Velocity in the Diffuse Crack Framework

Let $\boldsymbol{v}(\boldsymbol{x})$ be the continuous velocity field and let $\boldsymbol{n}$ be the unit normal pointing from the "−" side to the "+" side of the diffuse band. Place two material points a distance $\pm\ell_0/2$ from the nominal crack mid-surface along $\boldsymbol{n}$; their positions are

$$\boldsymbol{x}^{\pm} = \boldsymbol{x} \pm \frac{\ell_0}{2}\,\boldsymbol{n}. \tag{B.1}$$

Taylor-expand $\boldsymbol{v}$ about $\boldsymbol{x}$ at these two points (first order):

$$\boldsymbol{v}(\boldsymbol{x}^{\pm}) = \boldsymbol{v}(\boldsymbol{x}) \pm \frac{\ell_0}{2}\,\boldsymbol{\nabla}\boldsymbol{v}(\boldsymbol{x})\cdot\boldsymbol{n} + \mathcal{O}(\ell_0^2). \tag{B.2}$$

Subtracting gives the velocity jump across the band:

$$[\![\boldsymbol{v}]\!] \equiv \boldsymbol{v}^+ - \boldsymbol{v}^- = \boldsymbol{v}(\boldsymbol{x}^+) - \boldsymbol{v}(\boldsymbol{x}^-) = \ell_0\,\boldsymbol{\nabla}\boldsymbol{v}(\boldsymbol{x})\cdot\boldsymbol{n} + \mathcal{O}(\ell_0^2). \tag{B.3}$$

Written in index form and neglecting higher-order terms it becomes

$$[\![v_i]\!] = v_i^+ - v_i^- \approx \ell_0\,v_{i,j}\,n_j, \tag{B.4}$$

and the corresponding tangential slip rate becomes

$$v_s \approx \ell_0\ \|(\boldsymbol{\nabla}\boldsymbol{v}\cdot\boldsymbol{n})_{\tan}\|\,. \tag{B.5}$$

C Strain Energy Decomposition Schemes

In the phase-field fracture literature, two strain energy decomposition schemes are widely adopted and are briefly summarised here for completeness.

Spectral (Miehe) Decomposition

The spectral decomposition proposed by Miehe et al. (2) splits the strain energy into tensile and compressive parts based on the sign of the principal strains. The strain tensor is diagonalised as $\boldsymbol{\varepsilon} = \sum_i \varepsilon_i\,\boldsymbol{n}_i\otimes\boldsymbol{n}_i$, and the positive and negative parts are defined as

$$\boldsymbol{\varepsilon}^{\pm} = \sum_i \langle\varepsilon_i\rangle_{\pm}\,\boldsymbol{n}_i\otimes\boldsymbol{n}_i, \tag{7}$$

where $\langle x\rangle_{\pm} = (x \pm |x|)/2$ are the Macaulay brackets and $\boldsymbol{n}_i$ are the principal strain directions. The tensile and compressive strain energy densities are then

$$\psi^{\pm} = \frac{\lambda}{2}\big\langle \mathrm{tr}(\boldsymbol{\varepsilon})\big\rangle_{\pm}^2 + \mu\,(\boldsymbol{\varepsilon}^{\pm}:\boldsymbol{\varepsilon}^{\pm}), \tag{8}$$

and only ψ^+ is degraded by the phase field. This decomposition is most appropriate for predominantly tensile or Mode I loading, and is not well-suited to shear-dominated or mixed-mode conditions.

Volumetric–Deviatoric (Amor) Decomposition

The decomposition proposed by Amor et al. (4) follows the hydrostatic–deviatoric split of the strain energy. The tensile and compressive energy contributions are given by

$$\psi_0^+ = \frac{1}{2}K\langle \mathrm{tr}(\boldsymbol{\varepsilon})\rangle_+^2 + \mu\,(\boldsymbol{\varepsilon}' : \boldsymbol{\varepsilon}'), \tag{9}$$

$$\psi_0^- = \frac{1}{2}K\langle \mathrm{tr}(\boldsymbol{\varepsilon})\rangle_-^2, \tag{10}$$

where K is the bulk modulus, μ is the shear modulus, and $\boldsymbol{\varepsilon}' = \boldsymbol{\varepsilon} - \frac{1}{3}\,\mathrm{tr}(\boldsymbol{\varepsilon})\,\mathbf{I}$ is the deviatoric strain tensor. In this formulation, the entire deviatoric strain energy drives fracture regardless of the sign of the deviatoric principal strains, which can lead to unphysical crack growth under compressive shear in heterogeneous composites.

A comparison of the two schemes together with the hybrid decomposition adopted in the present work is given in Table 2.

Table 2: Comparison of strain energy decomposition schemes.

	Miehe (2)	Amor (4)	Present hybrid
Split basis	Principal strains	Volumetric–deviatoric	Volumetric + principal deviatoric
Compressive shear drives $\mathcal{H}$?	No	Indirectly yes	No
Degradation scheme	Anisotropic	Isotropic	Anisotropic
Suitable for	Mode I	Shear/mixed	Heterogeneous composites

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